

Macaque MERscope Ultra initial data analysis

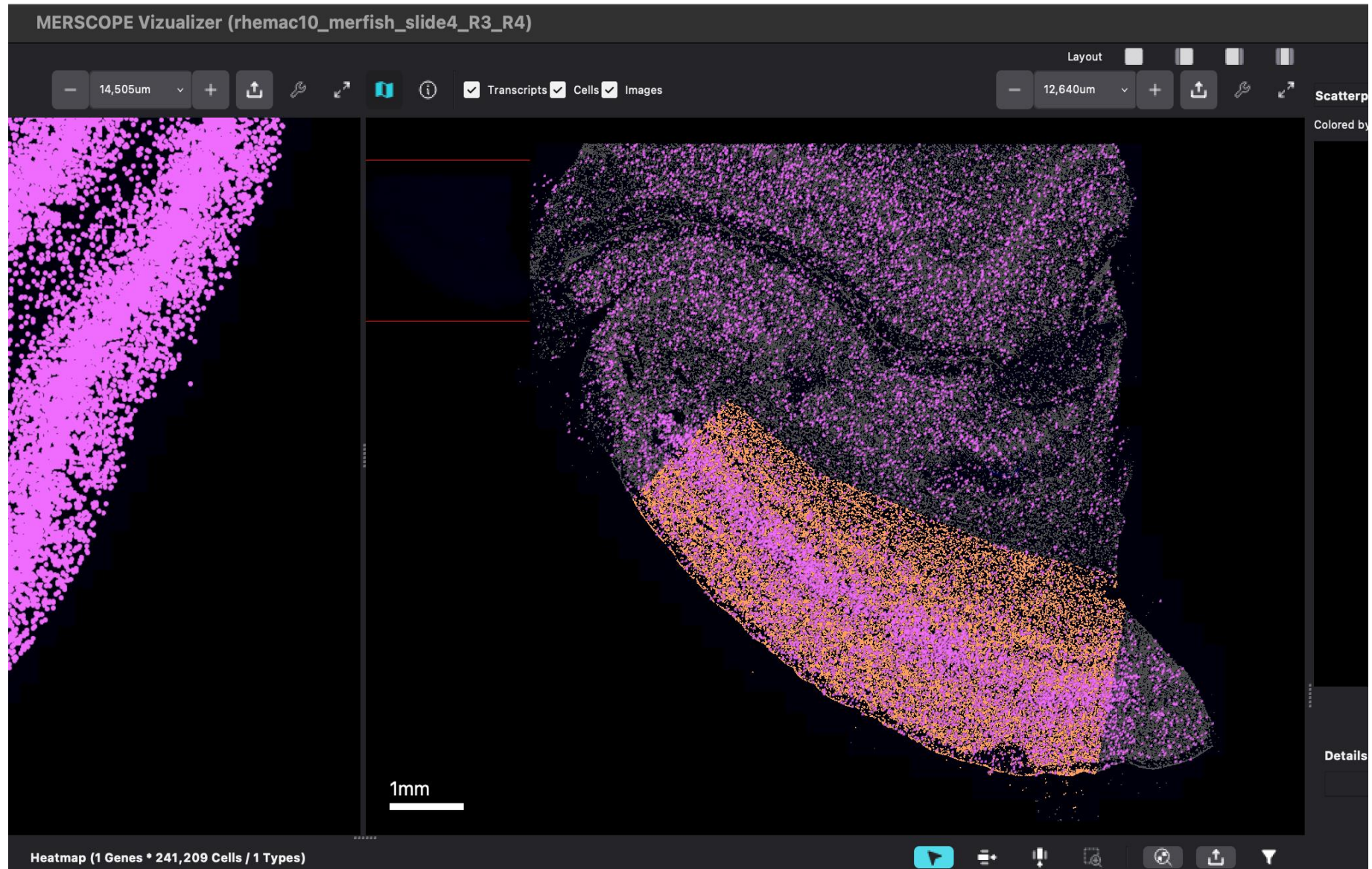
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Bioinformatician UMMS

5/8/2025

Vizualizer example slide4 R3 OSTN

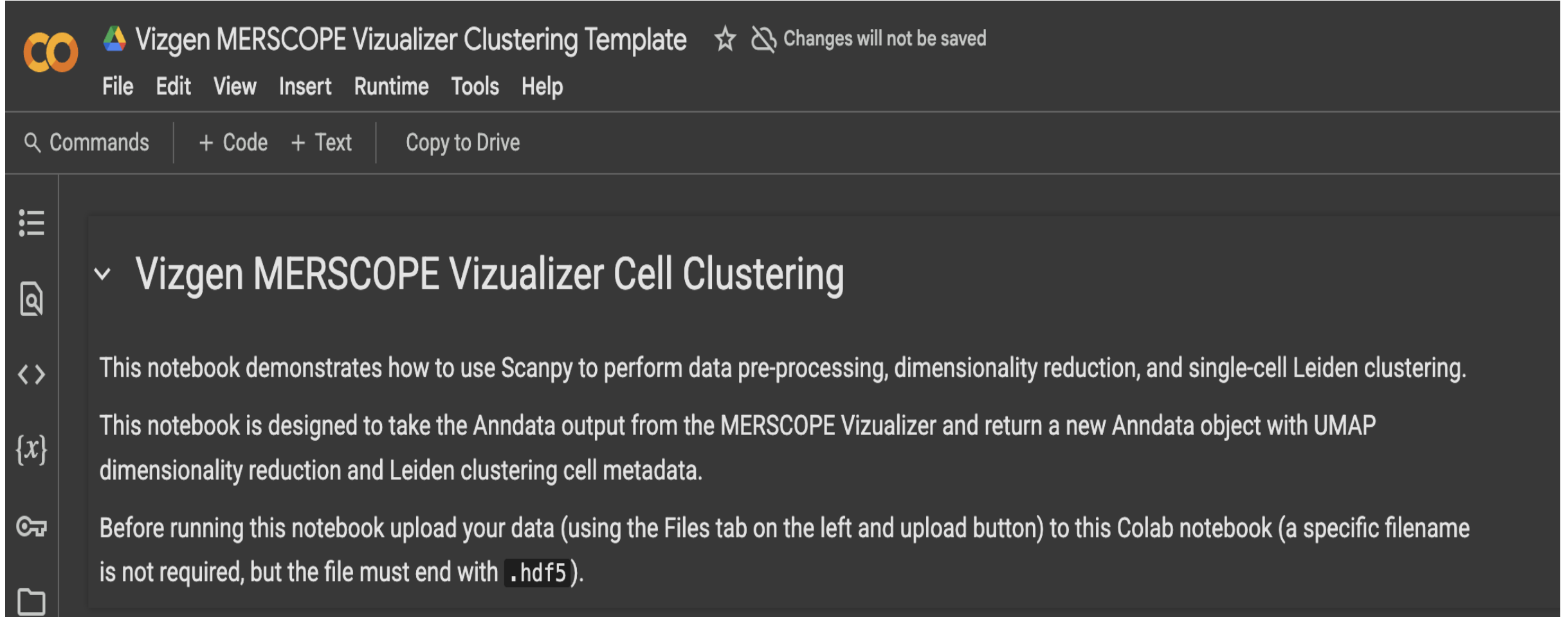
Pink = OSTN transcripts
Orange = Nuclei



Downstream analysis software

- Python
 - Vizgen officially supports
 - Scanpy
 - Used for scRNAseq/snRNAseq analysis (performs data pre-processing, dimensionality reduction, single-cell Leiden clustering)
 - Also performs analysis & visualization of spatial transcriptomics data
 - Squidpy
 - Used specifically for analysis & visualization of spatial transcriptomics data
- R
 - Currently, Vizgen doesn't officially support Seurat analysis yet
 - Seurat
 - Introduced extension October 2023 to analyze spatially-resolved RNA-seq data

Python analysis - Scanpy



The screenshot shows a Google Colab notebook interface. At the top, the title bar reads 'Vizgen MERSCOPE Vizualizer Clustering Template' with a star icon and a warning 'Changes will not be saved'. Below the title bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar labeled 'Commands' is followed by '+ Code' and '+ Text' buttons, and a 'Copy to Drive' button. On the left side, there is a sidebar with icons for a menu, search, navigation, code editor, and files. The main content area displays the notebook title 'Vizgen MERSCOPE Vizualizer Cell Clustering' with a dropdown arrow. Below the title, there are three paragraphs of text: 'This notebook demonstrates how to use Scanpy to perform data pre-processing, dimensionality reduction, and single-cell Leiden clustering.', 'This notebook is designed to take the Anndata output from the MERSCOPE Vizualizer and return a new Anndata object with UMAP dimensionality reduction and Leiden clustering cell metadata.', and 'Before running this notebook upload your data (using the Files tab on the left and upload button) to this Colab notebook (a specific filename is not required, but the file must end with `.hdf5`).'.

co Vizgen MERSCOPE Vizualizer Clustering Template ☆ ⚠ Changes will not be saved

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✓ Vizgen MERSCOPE Vizualizer Cell Clustering

This notebook demonstrates how to use Scanpy to perform data pre-processing, dimensionality reduction, and single-cell Leiden clustering.

This notebook is designed to take the Anndata output from the MERSCOPE Vizualizer and return a new Anndata object with UMAP dimensionality reduction and Leiden clustering cell metadata.

Before running this notebook upload your data (using the Files tab on the left and upload button) to this Colab notebook (a specific filename is not required, but the file must end with `.hdf5`).

Load Anndata object

- `ad_viz = sc.read_h5ad("09-04-25_18-18_rhemac10_merfish_slide4_R3_Cells.hdf5")`

View of AnnData object with `n_obs × n_vars = 241209 × 881`

obs: 'Custom Cells groups', 'Datasets', 'volume', 'center_x', 'center_y'

uns: 'Custom Cells groups_colors', 'Datasets_colors'

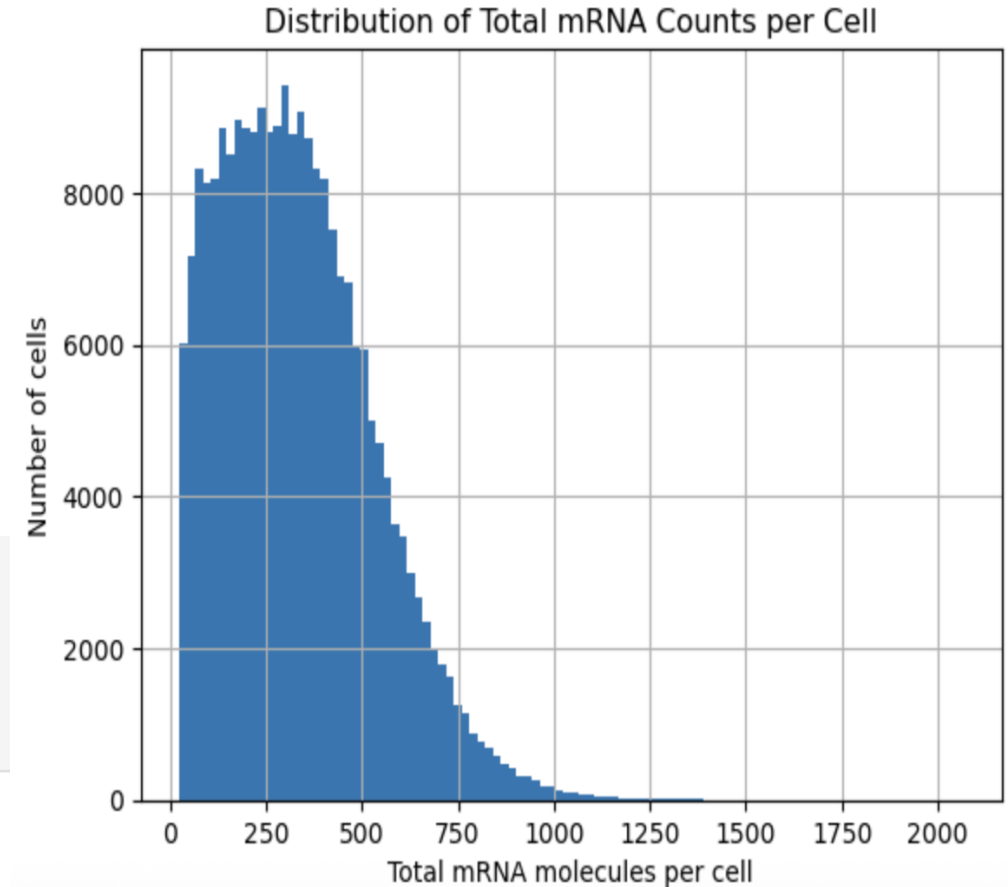
Filter Cells Based on Minimum Gene Expression Counts

- Filter the object by cells based on minimum gene expression
 - Keep cells with 25 mRNA transcripts or greater
- Cells with low total gene expression counts may indicate poor probe hybridization, low RNA capture efficiency, or other technical issues. Filtering out these cells is crucial to ensure that downstream analyses are based on reliable data

```
ad_viz = ad_viz[keep_cells]
# print object
ad_viz
# number of cells went from 241,209 to 236,965
```

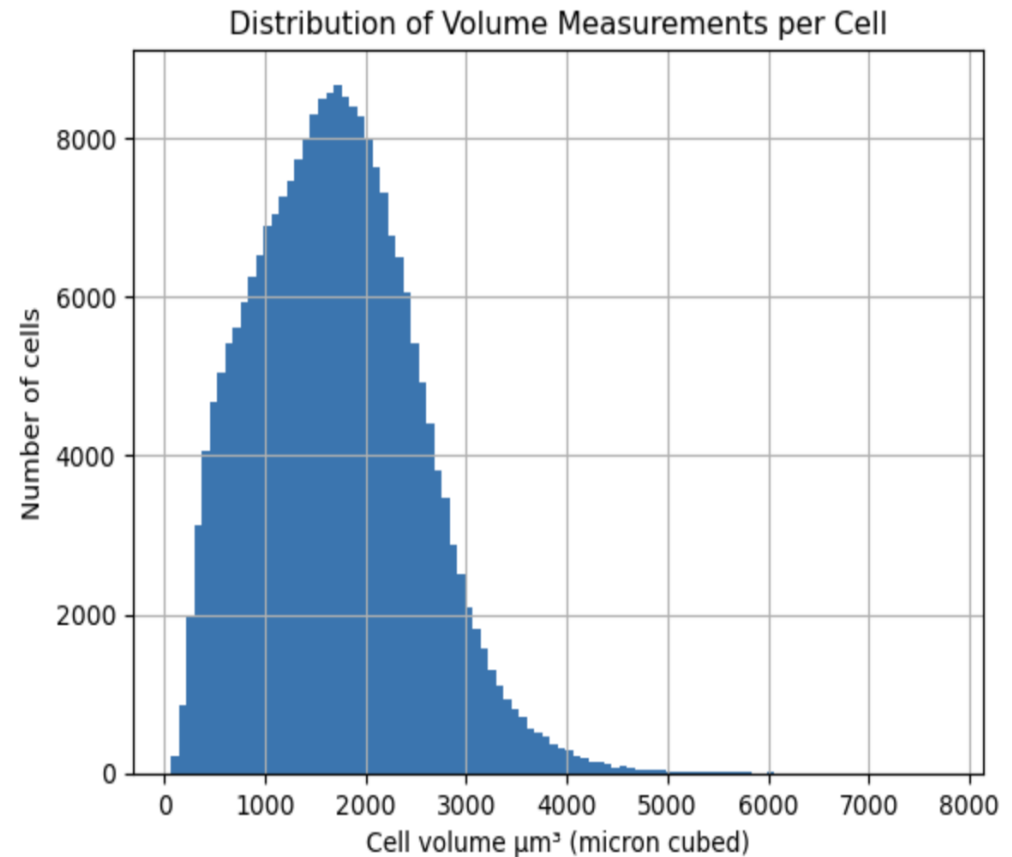
View of AnnData object with $n_{\text{obs}} \times n_{\text{vars}} = 236965 \times 881$
obs: 'Custom Cells groups', 'Datasets', 'volume', 'center_x', 'center_y'
uns: 'Custom Cells groups_colors', 'Datasets_colors'

```
241209-236965
# 4,244 cells removed
```



Filter Cells Based on Volume

- Cell volume is estimated physical size of each segmented cell in 3D space, usually measured in cubic microns (μm^3)
- Filter cells based on volume
 - Keep cells that have volume of 50 or greater
- Comes from the cell segmentation step, where each cell is outlined in space based on detected transcripts and imaging data (vizualizer)
- Used to filter out poorly segmented cells, debris, or cells that are too small to be biologically meaningful
- Didn't remove any cells from ad_viz object



UMAP and Single-Cell Clustering

```
# Normalize each cell by total counts over all genes, so that every cell has the same total count after normalization
# pp means pre-processing
sc.pp.normalize_total(ad_viz)

# apply log transformation to normalized data
# log(X + 1) where X is the data matrix of shape n_obs(cells) x n_vars (genes)
sc.pp.log1p(ad_viz)

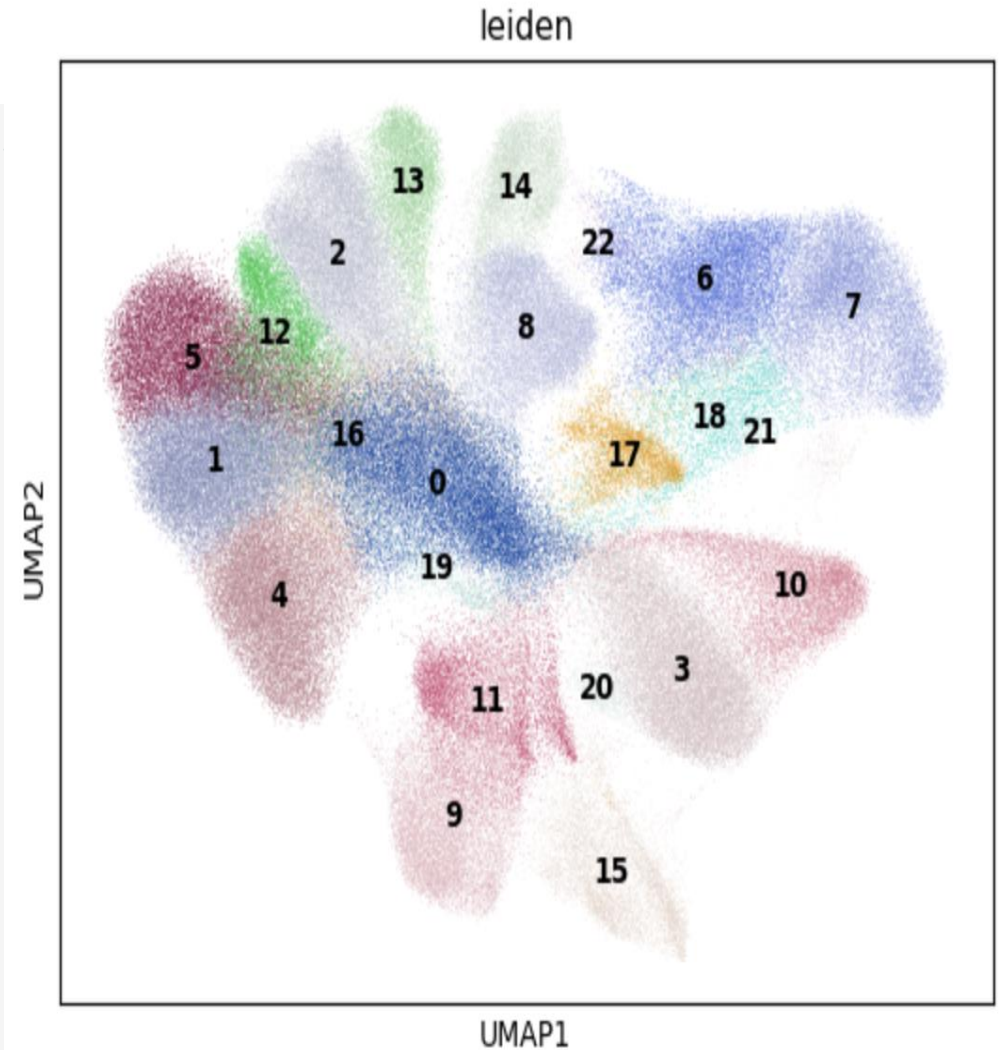
# Scale data to unit variance and zero mean
# max_value=10 caps extreme values to avoid distortion from outliers
sc.pp.scale(ad_viz, max_value=10)

# perform PCA on sparse matrix
# tl means tool
sc.tl.pca(ad_viz, svd_solver='arpack')

# find neighbors of cells (like the FindNeighbors function in Seurat!)
sc.pp.neighbors(ad_viz, n_neighbors=10, n_pcs=20)

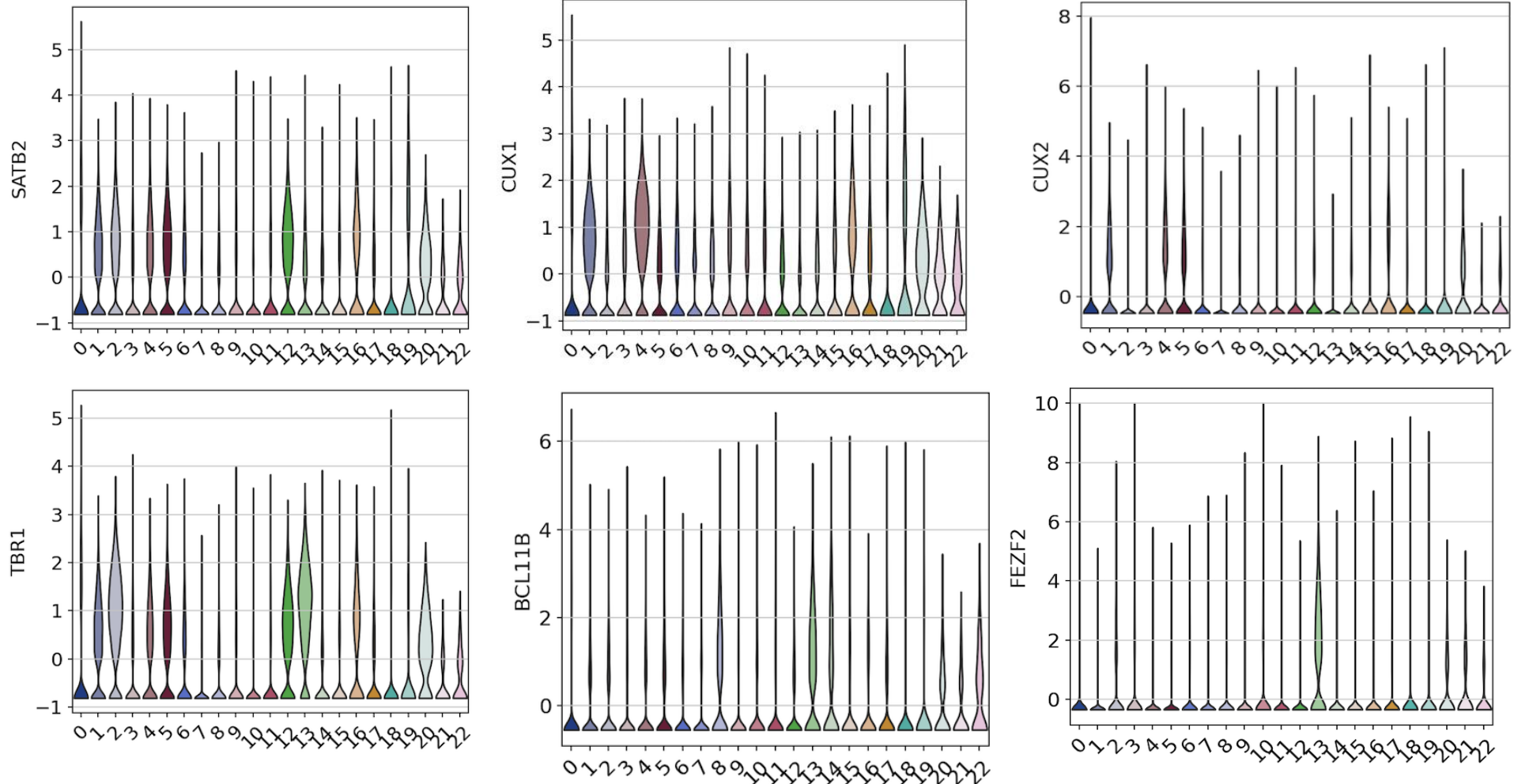
# computes UMAP embedding for visualization
# based on neighborhood graph calculated above
sc.tl.umap(ad_viz)

# performs leiden clustering based on the resolution defined with = above
# Cluster cells using the Leiden algorithm [Traag et al., 2019], an improved version of the Louvain algorithm [Blondel]
sc.tl.leiden(ad_viz, resolution=resolution)
```

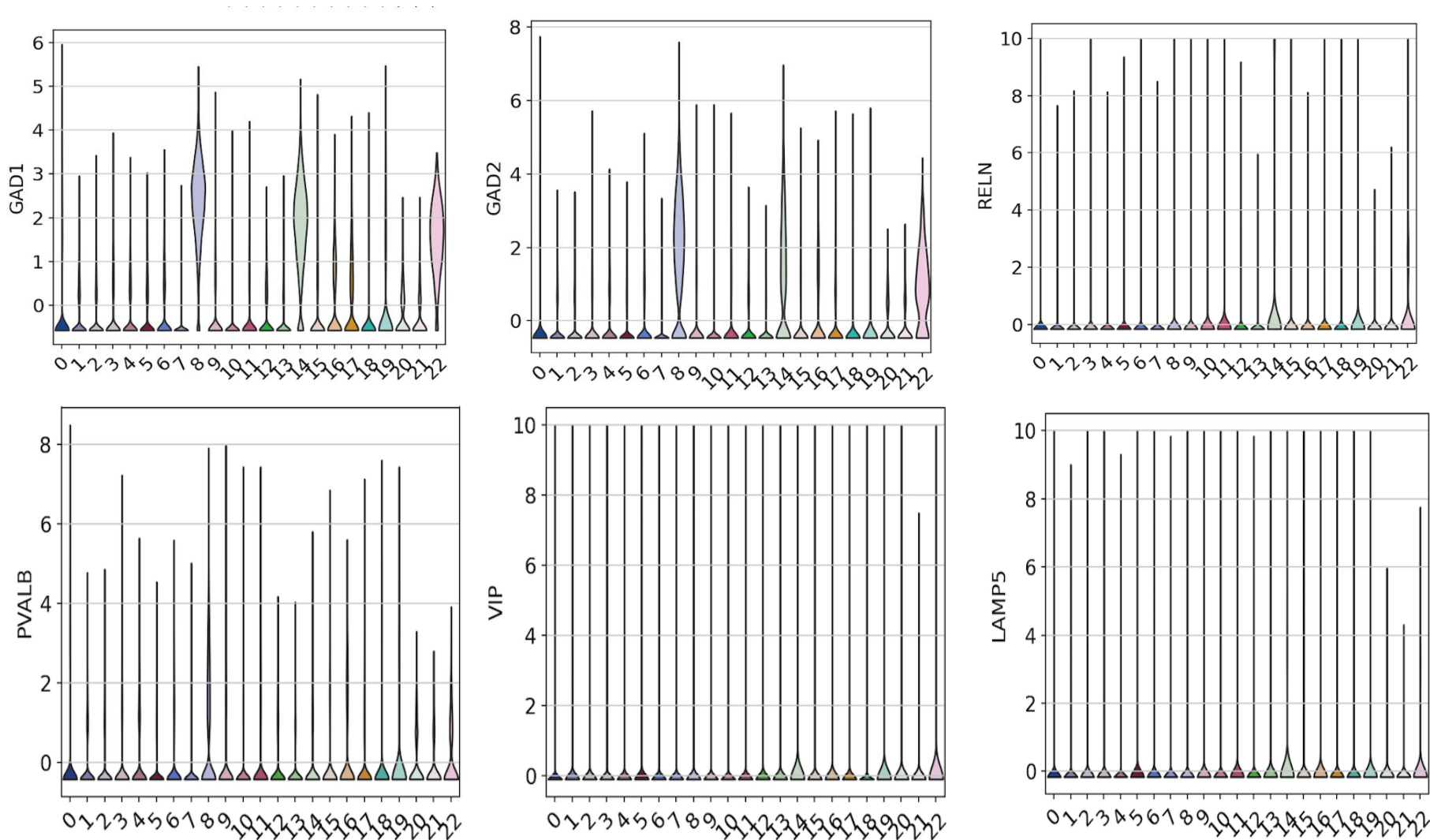


- PCs = 1-20
- Resolution = 1.5

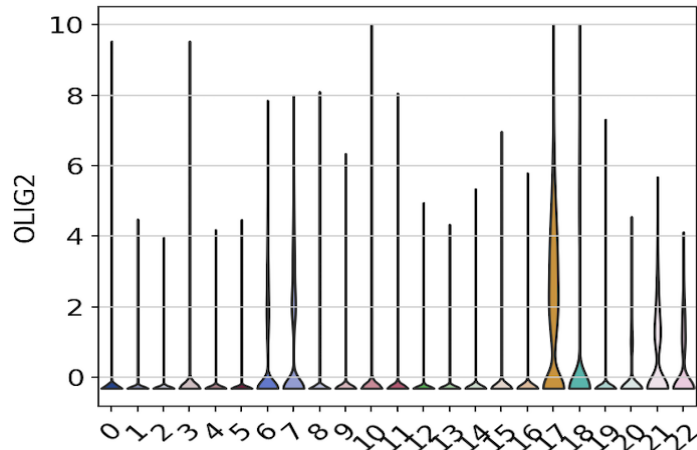
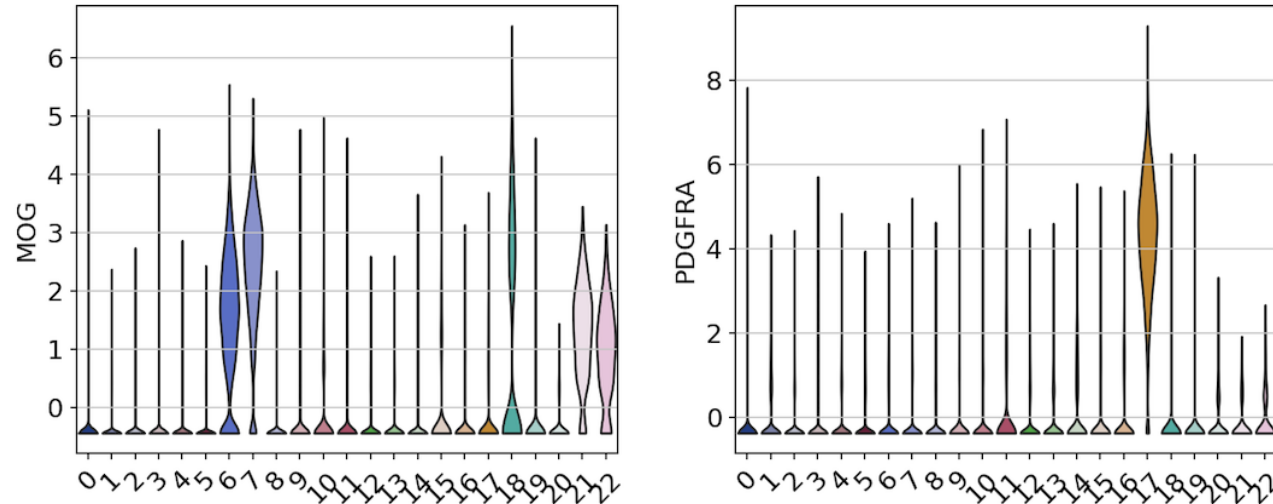
Gene violin plots – Excitatory clusters?



Gene violin plots – Inhibitory clusters?

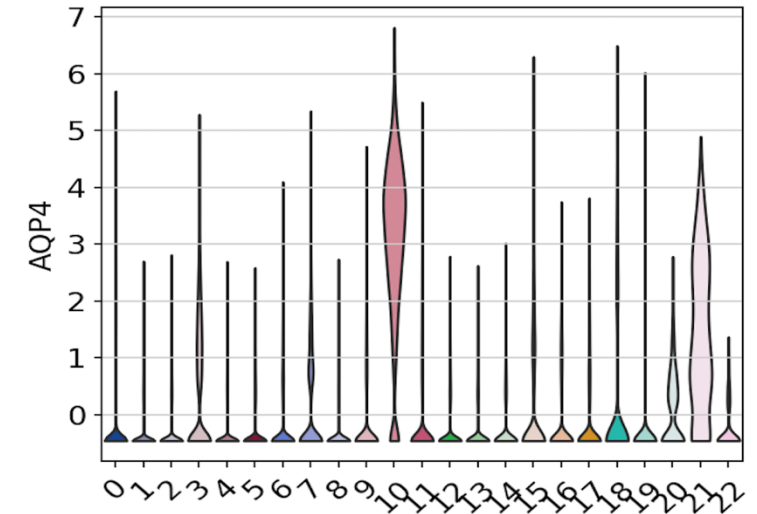


Gene violin plots – other ?

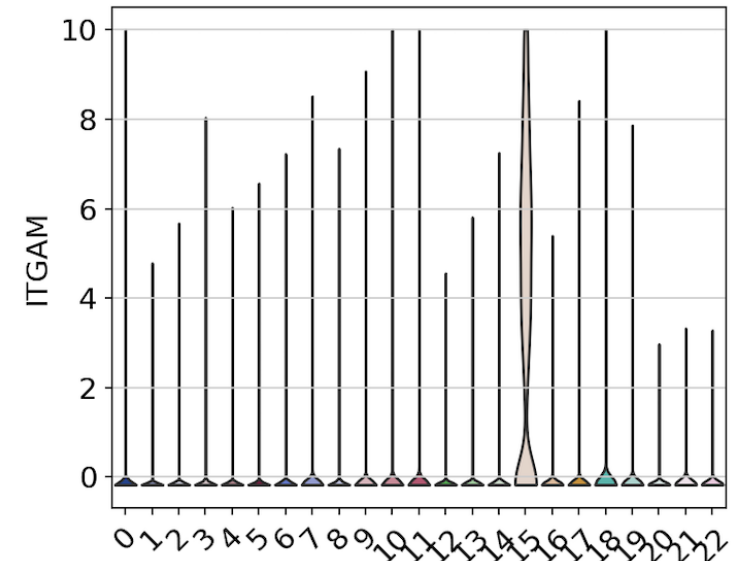


Oligodendrocytes

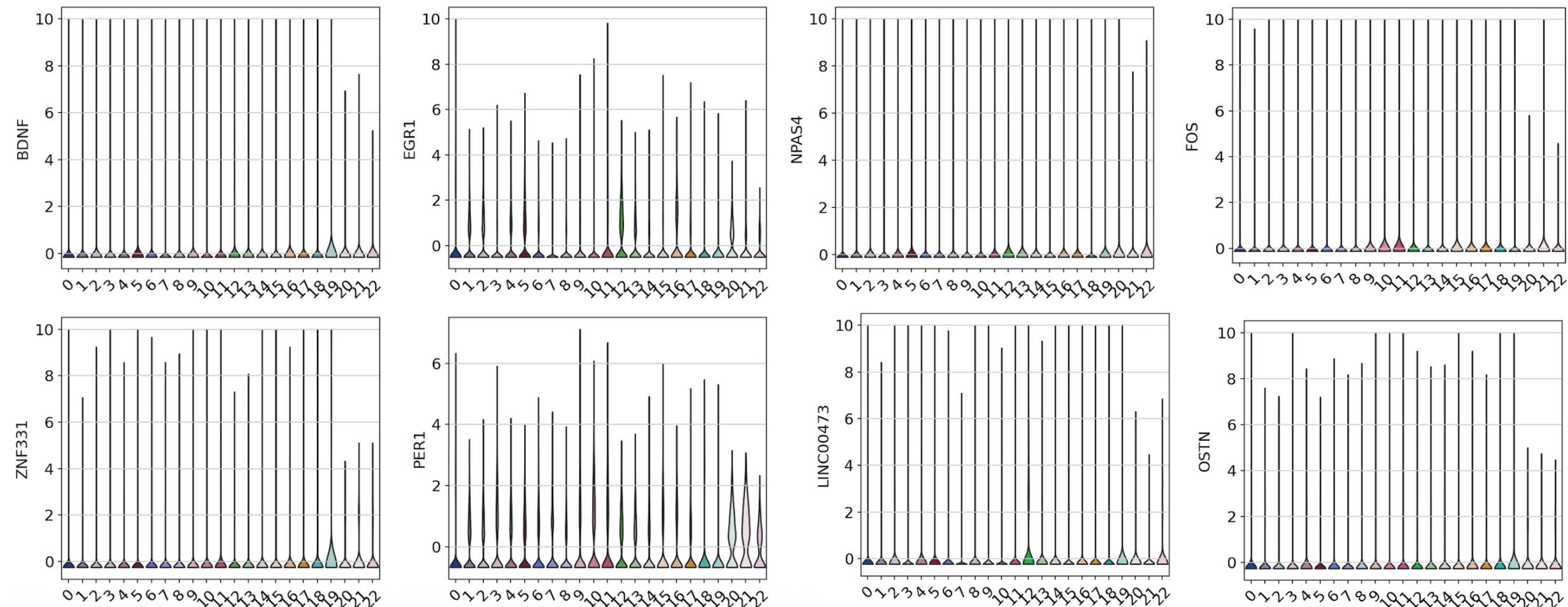
Astrocytes



Microglia



Gene violin plots – Activity Dependent Genes



Start annotating UMAP

